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Health Policy Analysis

Poor Patient Care Outcomes and Nurse Job Outcomes Associated With Unfavorable Intensive Care Unit and Emergency Department Nurse Work Environments: Implications for Critical Care Medicine

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ABSTRACT

Objectives: Efforts to improve critical care outcomes are traditionally focused on intensive care unit (ICU) work environments, despite the reality that nurses in emergency departments (EDs) also deliver critical care. EDs and ICUs in the same hospitals tend to be differently resourced and may have different work environments as assessed by nurses. The objective of this study was to assess similarities in ED and ICU nurse work environment evaluations and associations with patient care and nurse job outcomes.

Methods: Cross-sectional evaluation of ED and ICU nurses in 169 hospitals from a study of nurses licensed to work in New York and Illinois hospitals in the United States, the 2021 RN4CAST-New York/Illinois (NY/IL) survey, was administered electronically. K-means clustering classified hospitals into profiles on the basis of similarities in ED and ICU nurse work environment reports. Hospital-level regression models determined the association between the profiles and the following hospital-level outcomes, namely, patient care quality and safety, nurse burnout, job dissatisfaction, and intent to leave.

Results: Three hospital profiles characterized similarities and differences in nurses' favorable and unfavorable work environments: "ED and ICU nurse-favorable" (n = 67 hospitals), and "ED and ICU nurse-unfavorable" (n = 42); and "ED nurse–unfavorable" (n = 60) indicating less favorable environments for ED than ICU nurses. Hospitals that were unfavorable for both ED and ICU nurses, or unfavorable for ED nurses only were associated with higher percentages of poorer outcomes, as compared to hospitals in which nurses in both settings reported favorable environments.

Conclusions: To optimize critical care, better nurse work environments are needed in both ICUs and EDs.

Keywords: critical care, emergency department, intensive care unit, nurse, value.

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Highlights

- Despite the intensive care unit (ICU) and emergency department (ED) nurses sharing a common patient population, previous studies have only evaluated their work environments separately.
- This observational study used a K-means algorithm to classify hospital ED and ICU nurse work environments and to determine associations with patient care and nurse job outcomes.
- Higher percentages of poor outcomes were associated with hospitals' work environments that were unfavorable for both ED and ICU nurses, and ED nurses only.

Introduction

Hospital emergency departments (EDs) and intensive care units (ICUs), while distinct acute care settings, are interconnected in that they both provide care to critically ill patients.[1] Since the 1950s, tremendous advancements and financial investments in critical care[2] have been made through the development of hospital ICUs, where complex, high-cost care is delivered by highly specialized nurses. In addition to ICUs, critical care is also delivered in hospital EDs, where nurses are trained to stabilize and manage critically ill patients before being transferred to a dedicated critical care unit. Care of the critically ill is expensive for patients, payers, and healthcare organizations regardless of setting. Insurers in the United States spend, on average, approximately $80 000 per ICU hospitalization,[3] and approximately $30 000 per encounter for patients receiving critical care in the ED based on a sample of patient encounters in previous literature.[4]

The value of critical care delivery, defined as the quality of care per unit of cost,[5] has been evaluated in both the ICU and ED settings across the literature.[6-9] Evidence from economic models demonstrates the value of innovative critical care models, such as the implementation of ICUs within hospital EDs,[4,10] as well as the burdens of inefficient system constraints such as ICU patient boarding in EDs.[11] What is lacking in these cost analyses are aspects of human capital that facilitate high-quality care delivery to patients, such as the value of registered nurses for good patient outcomes.

Nurses are the most abundant professional group in healthcare[12] and have specialized training to provide surveillance, medication administration, and care escalation. Whether nurses are adequately supported in their work environments to deliver

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safe and quality critical care services is an important predictor of optimal outcomes for patients in EDs[13,14] and ICUs.[15,16] In more than 2 decades of research, the nurse work environment is conceptually defined as nurse staffing resource adequacy, physician-nurse collegiality, nurse involvement in hospital decision-making, nurse autonomy, and foundations for high-quality care.[17,18] The quality of ED and ICU nurses' work environments is associated with costly and avoidable patient outcomes such as hospital-acquired infections,[15,19] 30-day hospital mortality,[16] and bounce-back ED visits.[13] Hospital variation in the quality of nurses' work environments is also associated with high nurse burnout and turnover,[20,21] which are further costly to the US healthcare system.[22]

In recent years, the National Academy of Medicine,[23,24] the Society for Critical Care Medicine,[25] and the American Association of Critical Care Nurses have published strategic priorities centered on improving the working conditions for clinicians in stressful, high-stakes critical care settings. Since outcomes for critical care patients rely on care transitions from EDs and ICUs, nurses in these settings require a supportive work environment to advance high-quality care and incorporate new health technologies and interventions for this population. In addition, innovations and evidence-based interventions in critical care work environments are largely implemented by frontline nurses in EDs and ICUs.

Previous literature has focused on improving critical care delivery by modifying the ICU work environment, where most critical care is delivered. These efforts overlook the reality that the ICU patient trajectory is often influenced by their care experiences in EDs, where patients are stabilized and sometimes managed for extended periods, particularly if the ICU is at capacity. Although existing evidence demonstrates the prevalence of poor ED and (separately) ICU work environments, it is unknown whether ED and ICU nurses in the same hospitals have similar work environments, and whether variation in the 2 interrelated work environments is associated with patient and nurse outcomes. In other words, minimal research has evaluated ED and ICU work environments within the same hospital as a foundational feature of critical care delivery.

The objective of this observational study of nurses in the United States (New York and Illinois) was to determine the extent to which ED and ICU nurses within the same hospital have similar nurse work environment reports, and whether similarities are associated with patient care quality and safety, and nurse job outcomes (burnout, job dissatisfaction, intent to leave).

We hypothesized that 4 hospital profiles would be identified characterizing critical care nurse work environments ([1] ED and ICU nurse–favorable; [2] ED and ICU nurse–unfavorable; [3] ED nurse–unfavorable [ie, ED nurses rating less favorable work environments as compared to ICU nurses]; and [4] ICU nurse–unfavorable (ie, ICU nurses rating less favorable work environments as compared to ED nurses)). A secondary hypothesis was that ED nurse work environments would be less favorable than ICU nurse work environments and, therefore, would confer worse outcomes on the basis of previous literature, demonstrating that ED nurses report worse work environments and job outcomes as compared to nurses in other specialties.[26,27]

Methods

Theoretical Framework

The Nursing Human Capital Value model[28] is the theoretical framework informing this observational study. In this framework, nurses are considered crucial human capital to healthcare organizations because they provide necessary healthcare services to produce patient outcomes (eg, admissions, discharge, lower in-hospital mortality rates, care continuity). Due to nurses' proximity to the bedside and empirical evidence that nurse-reported outcomes are associated with objective patient outcomes[29,30]—nurses are valid informants of patient quality and safety.

A large body of evidence developed for more than 20 years[18,20,31-34] in the United States demonstrates a relationship between hospital nursing resources, such as the quality of nurses' work environments and the adequacy of staffing resources in hospitals, is associated with poor patient care quality and safety, and avoidable outcomes such as hospital readmissions and in-hospital deaths. Nurses spend the most direct care hours with patients as compared to other healthcare professionals; thus, the conditions in which they work significantly impact outcomes for patients, including the quality and safety of care delivery.

An investment in nursing services through resources such as safe patient-to-nurse staffing ratios[35] and high-quality work environments[36] is considered a structural foundation for healthcare delivery. This allocation of nursing care can contribute to cost reduction through the delivery of patient care and/or the leadership and decision-making of nurses that advances efficiency and value in healthcare.[28] In this study, we posit that the allocation of resources for ED and ICU nurses both influences outcomes for patients and nurses.

Study Design and Data Collection

The data for this observational study are from the RN4CAST-New York/Illinois (NY/IL) study of registered nurses conducted between April and June 2021. The methodology of the RN4CAST-NY/IL study has been published in detail elsewhere.[37,38] Nurses licensed to work in New York and Illinois were emailed an electronic survey in collaboration with the National Council of State Boards of Nursing. Nurses reported the name of their employer and demographics and answered questions about the quality of their work environment (eg, adequacy of nursing resources, presence of supportive nursing leadership), unit staffing levels, and their job outcomes (eg, burnout, job dissatisfaction). Nonresponders received regular follow-up reminders. The survey took 10 to 15 minutes to complete. At survey completion, 99% of acute care hospitals in New York and Illinois were represented.[37,39]

The sample had a mean of 5.8 ED nurses and 11.7 ICU nurses per hospital. The final sample comprised 2966 nurses (n = 978 ED; n = 1988 ICU nurses) across 169 hospitals. For context, in the United States, there are approximately 167 000 ED nurses[40] and 63 449 ICU nurses.[41] The RN4CAST-NY/IL survey purposively sampled nurses licensed in New York and Illinois and, thus, is not nationally representative of ED and ICU nurses in the United States.

Hospital Characteristics

Nurse survey data were linked with the American Hospital Association Annual Hospital Survey using a common hospital identifier.[42] This linkage allowed adjustment for hospital characteristics, including size (based on bed count), technology capabilities, teaching status, trauma center designation, and annual ED patient volume. Hospital size was categorized as small (≤250 beds), medium (251-500 beds), and large (>500 beds). Teaching status classifications included nonteaching (no residents), minor (1:4 resident-to-bed ratio), and major (>1:4 ratio). Hospital technology capability was defined as the provision of organ transplant services and open-heart surgery, whereas trauma designation was identified for hospitals with a certified trauma center. Annual ED patient volume was grouped into low (<40 000 patients/year), medium (40 000-80 000), and high (>80 000).

Measures

All exposure and outcome variables, as well as nurse demographics, were derived from the RN4CAST-NY/IL survey.[37,38]

Work Environment

The exposure variable for work environment quality was measured with a single-item question asking nurses to "rate the overall quality of your work environment," with response options ranging from excellent to poor. For analysis, the responses "good" and "excellent" were dichotomized into "good" and "poor," and "fair" was categorized into the "poor" category. The dichotomized responses from nurses were then aggregated to the hospital level to represent the proportion of ED and ICU nurses rating their work environment as poor.

Patient Outcomes

Patient outcomes included ratings of patient care quality and patient safety grades. Nurses assessed the quality of patient care on their unit using a 4-point Likert scale, which was then dichotomized into "favorable" and "unfavorable" quality of care. Nurses also provided a patient safety grade ranging from A to F (similar to a grading system dichotomized as "favorable" [A, B] and "unfavorable" [C, D, F]).

Nurse Job Outcomes

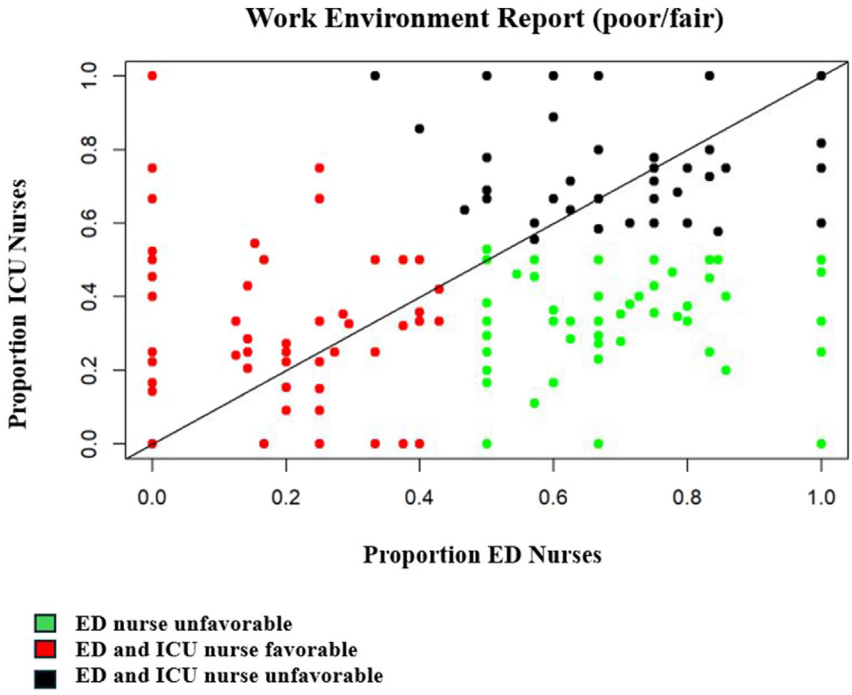
Nurse job outcomes included burnout, job dissatisfaction, and intent to leave. Burnout was measured using the Maslach Burnout Inventory emotional exhaustion subscale, with scores over 27[43] indicating "high burnout." Clinicians responded to a single-item question asking about job dissatisfaction using a 4-level Likert scale ranging from "very satisfied" to "very dissatisfied."[38,44] Finally, clinicians responded to a single-item question about whether they intend to leave their hospital in the next year. Measures were dichotomized for the data analysis, indicating high burnout, job dissatisfaction, and intent to leave the job.

Data Analysis

Profiles of hospitals based on how similarly ED and ICU nurses report on their work environment were identified via the K-means clustering algorithm. The K-means algorithm was selected to easily explore whether hospitals would cluster with respect to 2 continuous measures: the proportions of ED and ICU nurses within a hospital rating their work environment as fair/poor. K-means tends to work better than latent class models and hierarchical clustering when the measures used are continuous; it is computationally efficient and simpler than latent class models. The drawback of K-means is the need to identify the optimal number of clusters K. However, the NbClust[45] package in R (the R Foundation for Statistical Computing, Vienna, Austria) uses 23 different indices/criteria to determine the optimal number of clusters. We fitted K-means with K ranging from 2 to 10, and 8 of the indexes proposed 3 as the best number of clusters. K equals 2 was the second best with 6 indices supporting it. K-means is a clustering algorithm that minimizes the within-cluster variance by assigning each data point to the nearest cluster center. Therefore, each point is assigned to the cluster that minimizes the distance (Euclidean distance in this case) between the point and the cluster center. The 3 cluster centers were the following: c1 equals (0.31,0.19); c2 equals (0.75, 0.71); and c3 equals (0.33, 0.69).

We produced a graph (Fig. 1) depicting each hospital as a point in the 2-dimensional space, in which the horizontal axis is the

Figure 1. Hospital profiles based on emergency department and intensive care unit nurse work environment reports (n = 169 hospitals).



Note. Each color represents a hospital profile based on ED and ICU nurses' report on work environment.

ED indicates emergency department; ICU, intensive care unit.

proportion of ED nurses, and the vertical axis is the proportion of ICU nurses reporting a poor work environment, and each hospital's membership in the identified profiles.

Student's *t* tests and analysis of variance tests at the hospital-level were used to determine differences in nurse demographics and percentages of poor/fair work environment reports, and poor patient care and nurse job outcomes between ED and ICU nurses. The same tests were used to test differences in hospital characteristics across hospital profiles. Hospital-level linear regression models were fitted to determine the relationship between the hospital profiles (exposure variable) and the patient care and nurse job outcomes, adjusted and unadjusted for hospital characteristics. We also fitted separate models for ED and ICU nurses. We prespecified the omitted category as the hospital work environment, where both ED and ICU nurses reported a favorable work environment.

Results

The final sample included 2966 nurses (n = 978 ED; n = 1988 ICU nurses) employed in 169 study hospitals (Table 1). There were no differences in the average hospital nurse age or self-reported gender between ED and ICU nurses. Years of experience in the hospital were on average higher among ICU nurses as compared to ED nurses (11 vs 9 years, $P < .001$). There were higher percentages of ICU nurses who self-reported as Asian race as compared to ED nurses (12% vs 8%, $P < .010$), but higher percentages of nurses who identified as "other" race in EDs (6% vs 3%, $P < .050$). Close to half of all clinicians (46%) reported a poor or fair work environment, and higher percentages of ED compared to ICU nurses reported a poor or fair work environment (50% vs 43%, $P < .050$).

The K-means algorithm identified 3 hospital profiles shown in Figure 1. Two hospital profiles were characterized by ED and ICU nurses agreeing on their work environment report as either unfavorable ("ED and ICU nurse–unfavorable", n = 42) or favorable ("ED and ICU nurse–favorable," n = 67). The third profile included hospitals with ED nurses rating the work environment less favorably than ICU nurses ("ED nurse–unfavorable," n = 60). No hospital profiles indicated that ICU nurses rated their work environment less favorably than ED nurses. Characteristics of the 169 study hospitals across profiles are listed in Table 2. There were no significant differences in hospital bed size, teaching status, technology capabilities, and trauma center designation across profiles.

Differences in study outcomes across the 3 hospital profiles are provided in Appendix Table 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.06.017. In the overall clinician models, on average, percentages/rates of nurses reporting poor patient care quality, unfavorable patient safety grades, high burnout, job dissatisfaction, and intent to leave were highest in hospitals where both ED and ICU nurses reported unfavorable work environments. The same outcomes were observed in the models for ICU nurses only. In the models with ED nurses only, higher percentages of nurses reported poor patient care quality and unfavorable safety grades in the hospitals, with ED and ICU nurses reporting an unfavorable environment. However, ED nurses reported the highest percentages of burnout and intent to leave in hospitals that were unfavorable for ED nurses only.

Findings from the adjusted hospital-level regression models are outlined in Table 3 for outcomes reported from all clinicians and also separately from ED and ICU nurses. The unadjusted models are provided in Appendix Table 2 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.06.017. The omitted category was hospitals with favorable work environments for both ED and ICU nurses (shown as "constant"). We provided a step-by-step description of how to interpret the findings with a focus on the magnitude of the associations:

Table 1. Hospital-level demographics, work environment reports, and outcomes reported by nurses in 169 hospitals.

Characteristics	All clinicians	ED RNs	ICU RNs	*P* value
Overall sample demographics				
Age (years), mean (SD)	43 (6)	43 (6)	43 (6)	.7
Female, % (SD)	86 (15)	84 (15)	87 (14)	.2
Years in hospital, mean (SD)	10 (5)	9 (5)	11 (6)	<.001
Ethnicity, % (SD)				
Hispanic	6 (10)	7 (12)	5 (8)	.1
Race, % (SD)*				.1
Asian	10 (16)	8 (14)	12 (17)	<.01
American Indian or Alaskan Native	0.9 (4)	0.8 (4)	0.9 (4)	.9
Black or African American	10 (17)	10 (17)	11 (17)	.7
Native Hawaiian or Other Pacific Islander	0.7 (3)	0.7 (3)	0.7 (4)	.9
Other	4 (9)	6 (12)	3 (6)	<.05
White	75 (28)	77 (28)	73 (28)	.3
Work environment, % (SD)				
Poor or fair environment	46 (28)	50 (30)	43 (26)	<.05
Patient outcomes, % (SD)				
Poor patient quality of care	28 (28)	32 (29)	24 (25)	<.05
Unfavorable patient safety grade	49 (32)	55 (33)	44 (30)	<.01
Job outcomes, % (SD)				
High burnout	54 (27)	55 (28)	53 (26)	.6
Job dissatisfaction	33 (25)	35 (27)	31 (23)	.2
Intent to leave	27 (24)	26 (25)	27 (24)	.5

Note. Table 1 represents the average outcomes for clinicians at the hospital level across the 169-hospital sample. P values indicate differences in hospital ED and ICU nurse demographics, work environment, patient outcomes, and nurse job outcomes.
ED indicates emergency department; ICU, intensive care unit; RN, registered nurse.
*Percentages will not equal 100 because participants could select more than 1 category. Within the hospitals, the clinician sample included 978 ED and 1988 ICU nurses.

Table 2. Hospital characteristics in 169 study hospitals across hospital profiles.

Hospital characteristic	Hospital profile			*P*-value
	ED and ICU nurse favorable (n = 67)	ED and ICU nurse unfavorable (n = 42)	ED nurse unfavorable (n = 60)	
Hospital size, n (%)				.9
≤ 250	25 (38)	14 (33)	19 (32)	
251-500	29 (43)	20 (48)	27 (45)	
>500	13 (19)	8 (19)	14 (23)	
Annual ED patient volume, n (%)				.7
0-40 000	21 (31)	11 (26)	13 (22)	
40-80 000	25 (37)	15 (36)	27 (45)	
>80 000	21 (32)	16 (38)	20 (33)	
Teaching status, n (%)				.14
No	23 (34)	7 (17)	16 (27)	
Minor	21 (31)	20 (48)	17 (28)	
Major	23 (35)	15 (35)	27 (45)	
Technology status, n (%)				.8
Not High Tech	31 (46)	22 (52)	28 (47)	
High	36 (54)	20 (48)	32 (53)	
Trauma center, n (%)				.9
No	25 (37)	16 (38)	21 (35)	
Yes	42 (63)	26 (62)	39 (65)	

Note. Fisher's exact tests were performed for categorical variables (ED annual volume, teaching status, technology status, trauma center), and global ANOVA tests were performed for continuous variables (Avg. % of clinicians rating their work environment as poor).
ANOVA indicates analysis of variance; Avg., average; ED, emergency department; ICU, intensive care unit.

1. Compared to the hospitals belonging to the omitted category (favorable environments for both ED and ICU nurses) and while holding all the other covariates constant, hospitals with unfavorable environments for ED and ICU nurses had a burnout rate 16.9 percentage points higher (β = 16.9, 95% confidence interval [CI] 10.2-23.6, $P < .001$).
2. Compared to the omitted category, hospitals with unfavorable environments for ED nurses only (and while holding the other covariates constant) had a burnout rate 10.4 percentage points higher (β = 10.4, 95% CI 4.5-16.4, $P < .010$).

Similar relationships were observed for poor patient care quality, unfavorable hospital safety grades, and nursing job dissatisfaction. In the models evaluating outcomes of ED nurses only, the same relationships were identified as in the overall clinician models, with increased percentages of poor outcomes, on average, in hospitals with unfavorable ED and ICU nurse work environments, and hospitals with unfavorable ED nurse work environments, as compared to the omitted category. Hospitals with unfavorable environments for both ED and ICU nurses on average had a poor patient care quality rate 26.6 percentage point higher (β = 26.6, 95% CI 15.9-37.3, $P < .001$) than hospitals with favorable work environments for both ED and ICU nurses when holding all the other variables constant; hospitals with unfavorable work environments for ED nurses only had a poor patient care quality rate 22.4 percentage point higher (β = 22.4, 95% CI 12.8-31.9, $P < .001$).

The ICU nurse models were distinct from the ED nurse models in that the only hospital profile significantly different from the omitted one for all the outcomes was the one labeled "ED and ICU nurse work environment-unfavorable." There were no statistically significant differences between the outcomes reported by ICU nurses and hospitals with unfavorable ED nurse work environments.

A sensitivity analysis was conducted, adjusted for hospital-level case mix and the number of inpatient Medicaid facility days (see Appendix Table 3 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2025.06.017). Similar relationships were observed as compared to the Table 3 models.

Discussion

Three hospital profiles described ED and ICU nurses' work environments: ED and ICU nurse–favorable, ED and ICU nurse–unfavorable, and ED nurse–unfavorable. In most study models, we identified significantly worse patient care and nurse job outcomes in hospitals with unfavorable work environments for ED and ICU nurses, as well as in hospitals with unfavorable ED nurse work environments. A distinct exception was the models for ICU nurses only, in which hospitals with unfavorable ED work environments were not associated with significant differences in ICU patient and nursing job outcome reports. This finding may be attributed to ICUs being less influenced by ED work environments, whereas EDs are impacted by the operations of inpatient units (eg, hospital crowding, bed availability, number of daily discharges).

Two of the hospital profiles demonstrated a "concordance" in ED and ICU nurse work environment reports (favorable or unfavorable), which is supported by literature,[26,37] demonstrating that the broader context or work environment of the hospital may impact different healthcare units in similar ways. The third hospital profile was unfavorable for ED nurses only, which aligns with evidence demonstrating that ED nurses report worse work environments than nurses in other healthcare units.[26,27] This disparity in work environment quality is due to cultural factors (high-stress, fast-paced care) in EDs, as well as policies that influence nurses' workloads in the United States.[46] For example, the Emergency Medical Treatment and Labor Act[47] specifies that all patients

Table 3. Adjusted hospital-level linear regression models for emergency department and intensive care unit profiles and outcomes.

Outcomes	Clinician group		
	Overall (95% CI)	ED nurses (95% CI)	ICU nurses (95% CI)
Poor care quality			
Constant	14.4* (7.7, 21.1)	12.5† (1.4, 23.5)	13.9‡ (4.5, 23.2)
ED & ICU nurse-unfavorable	26.3* (19.9, 32.8)	26.6* (15.9, 37.3)	23.9* (14.9, 32.9)
ED nurse-unfavorable	8.0‡ (2.2, 13.8)	22.4* (12.8, 31.9)	−1.8 (−9.8, 6.3)
Unfavorable safety grade			
Constant	33.7* (25.9, 41.6)	33.0* (22.0, 44.1)	29.9* (19.3, 40.6)
ED & ICU nurse-unfavorable	33.7* (26.1, 41.3)	39.1* (28.4, 49.8)	32.9* (22.6, 43.2)
ED nurse-unfavorable	16.1* (9.3, 22.9)	37.4* (27.9, 46.9)	4.1 (−5.2, 13.3)
Burnout			
Constant	50.0* (43.1, 56.9)	44.3* (33.8, 54.8)	49.1* (38.9, 59.3)
ED & ICU nurse-unfavorable	16.9* (10.2, 23.6)	21.7* (11.5, 31.8)	13.5‡ (3.6, 23.3)
ED nurse-unfavorable	10.4‡ (4.5, 16.4)	23.5* (14.4, 32.5)	2.8 (−6.0, 11.0)
Job dissatisfaction			
Constant	28.1* (21.6, 34.6)	23.6* (13.9, 33.3)	27.9* (19.5, 36.5)
ED & ICU nurse-unfavorable	22.7* (16.4, 29.0)	25.9* (16.6, 35.4)	21.9* (13.7, 30.2)
ED nurse-unfavorable	10.9* (5.2, 16.6)	26.8* (18.4, 35.2)	−0.9 (−8.3, 6.5)
Intent to leave			
Constant	25.4*(18.3, 32.5)	20.6* (10.7, 30.6)	25.9* (20.3, 31.6)
ED & ICU nurse-unfavorable	11.7‡ (4.8, 18.5)	13.4‡ (3.8, 23.1)	11.5† (2.0, 20.9)
ED nurse-unfavorable	5.2 (−0.9, 11.4)	14.2‡ (5.6, 22.8)	−4.7 (−13.2, 3.7)

Note. The table reports the mean outcome for the "ED and ICU nurse favorable" omitted category (represented as "constant"), and the differences between the mean outcome for hospital profiles 2 (and 3) and the reference profile (agree, favorable environment).
CI indicates confidence interval; ED, emergency department; ICU, intensive care unit.
Significance levels:.
*$P < .001$.
†$P < .050$.
‡$P < .010$.

entering a hospital ED in the United States must be evaluated for care regardless of insurance status or hospital bed availability, which may impact nurse workload.

Overall, our findings are supported by a large evidence base demonstrating that unfavorable nurse work environments are associated with nurse-reported and objective patient outcomes.[26,48] Two decades of evidence demonstrate[49-52] the mechanisms of this underlying relationship. When nurses work in unfavorable work environments, core tasks that ensure high-quality patient outcomes are missed, such as medication administration, skin care, and communicating care plans.[52,53] Operational failures such as missing supplies or care orders and failures in the electronic health record are aspects of unfavorable nurse work environments associated with poor patient outcomes.[54]

Our study is novel in that we evaluated the context of care in 2 interrelated work environments that impact outcomes for critical care patients.[55] That unfavorable work environments in either the ED only or in the ED and ICU are associated with worse outcomes reported by ED nurses is concerning, given that hospital critical care is often initiated in hospital EDs, where critical care patients are stabilized and may even board[11,56] for hours before transfer to the ICU. The implication of this finding is that hospital investments in safer working conditions for nurses are needed not just in specialized care areas such as the ICU, but also in the ED.

Our findings extend existing research demonstrating that poor ICU and ED work environments are associated with worse patient outcomes in their respective healthcare settings. To our knowledge, this study is one of the first to evaluate the patient care and nurse job outcomes among nurses in both settings, given that critical care trajectories often begin in the ED and end in the ICU. We leverage clinician-reported outcomes in our study,[57,58] which are increasingly considered in health economic modeling, given the value that frontline healthcare professionals provide for patient outcomes and healthcare costs.[22,59,60]

In 60% of our study hospitals, nurses rated their critical care work environment as poor or fair. Based on previous literature, poor nurse work environments are characterized by high nurse workloads (high patient-to-nurse staffing ratios), unsupportive unit managers, and a lack of interprofessional collaboration that may not be conducive to the implementation of new technologies and interventions. Our findings identify that advancements in critical care depend on a high-quality nurse work environment, given that nurses oversee care delivery and the piloting of new interventions in their workplace. Therefore, future cost models evaluating the value of critical care interventions should capture the costs of investing in high-quality nurse work environments to adequately support piloting and implementing new care delivery models.

Specific strategies to improve the quality of nurses' work environments that are evidence-based include implementing safe patient-to-nurse ratios to improve nurse workloads[61] and addressing patient throughput issues[62] that influence the quality

of patient care. Hospital patient-to-nurse staffing ratios[33,48,63] are a key component of the nurse work environment that are consistently associated with patient outcomes and nurse job outcomes. Evidence demonstrates that every 1 patient added to a nurse's workload is associated with higher odds of patient mortality, hospital readmissions, prolonged length of stay, and higher odds of nurse burnout.[48,63]

Mandated nurse staffing ratios is a health policy aimed at improving nurses' work environments that has been successfully implemented in California in 2004 through HB 394.[32,61,64] Since implementation, evidence demonstrates that patients experience more direct care nursing hours in California[64] compared to other states, and nurses report lower burnout, job dissatisfaction, and intent to leave.[32] Evidence also demonstrates that nurses in California are less likely to work in an unfavorable work environment compared to nurses in states without a hospital nurse staffing mandate.[32]

In recent years, more states have introduced nurse staffing mandates given the important implications for patient outcomes and payer costs (eg, cost savings due to better patient outcomes). Oregon became the second state to enact hospitalwide nurse staffing mandates in 2023.[65] Although other states, such as Massachusetts, have instituted nurse staffing mandates in ICUs[66] only, our findings emphasize the importance of hospital-wide staffing legislation, given that critical care is delivered outside of ICUs. Although these policy interventions require hospital costs, economic models are needed to evaluate the value of such investments from an outcomes and cost perspective (eg, cost savings owing to avoidable readmissions and lower nurse turnover) through cost-effective analyses.

Implications for Future Health Economic Models and Health Policy

Nursing care influences existing value frameworks (The Value Flower[5,67]) in health economics directly and indirectly through the patient quality-adjusted life years gained, insurance, equity, productivity, and net costs. A substantial body of evidence demonstrates that hospitals with supportive nurse work environments experience better patient outcomes,[15,68-70] fewer costs for payers,[71,72] and fewer penalties to hospitals due to high-value care delivery.[73] When nurses operate in supportive work environments, they are less likely to experience burnout and turnover, which are costly to patients (through poor outcomes) and hospitals (through turnover).[36] As such, nurses provide value from a payer and societal perspective. Our study contributes to existing literature that nurses are a value driver in critical care; thus, economic models must quantify the investments in their work environment.

Just as existing cost models capture health utilities, future models could measure the proportion of nurses reporting high-quality or poor/fair work environments. Alternatively, models could include validated measures of nurses' work environments, such as the Practice Environment Scale-5, as a utility of nursing resources.[74] This measure is informative to the potential success of a health intervention or technology as well as the costs and consequences of patient and nurse outcomes (eg, cost savings due to better patient outcomes and more nurses remaining in a job). For example, implementing a new critical care technology may be unexpectedly more expensive than projected if, upon piloting, a large proportion of nurses leave their jobs, requiring a hospital to pay more in healthcare labor, onboarding, and orientation.

Limitations

Our study uses nurse-reported patient outcomes; however, empirical evidence in health services and health economics research[30,57,58] demonstrates that frontline clinician perspectives are both correlated with objective patient outcomes and inform the value of healthcare delivery. Nurses' reports on patient care quality and safety specifically are informed by their provision of care delivery and surveillance directly at the bedside and are empirically correlated with objective patient outcomes,[30] though the potential for reporting bias remains. Causal inferences cannot be drawn from our findings due to the observational nature of the study design. Finally, while the data were collected during the COVID-19 pandemic, we believe our findings remain generalizable; previous studies have shown consistency in nurses' assessments of their work environments both before and during the pandemic.[37,39]

Conclusions

Unfavorable hospital work environments, whether for both EDs and ICUs or EDs only, were associated with worse patient care and nursing job outcomes. To improve the value of critical care delivery, systems-based investments are needed in both the ED and ICU nurse work environments. Economic models must capture hospital investments in high-quality nurse work environments when measuring the value of emerging critical care interventions, given that better patient outcomes are contingent upon the nurse work environment.

Author Disclosures

Author disclosure forms can be accessed below in the Supplemental Material section.

Supplemental Material

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